Underwater Marine Debris Detection

Sonar Imagery · Deep Learning · Sensor Fusion

Project Overview & Methodology · June 2026

Framework	PyTorch 2.12 · YOLOv8 · SegFormer (HuggingFace)
Hardware	Apple M4 MPS GPU · Python 3.11 · macOS 15
Dataset	FLS Marine Debris â■■ Valdenegro-Toro et al., OCEANS 2025
Difficulty	Hard â■■ class imbalance, cluttered sonar, multi-modal fusion

1. Project Goal

Detect and classify underwater marine debris from Forward-Looking Sonar (FLS) imagery. Three task tracks: **sonar classification** of cropped object images, **sonar detection** using bounding-box annotations, and a **fusion strategy** combining cross-domain pretraining with fine-tuning. Optical-sonar comparison via TrashCan dataset is a future phase.

2. Problem Statement

Marine debris monitoring is critical for ocean health. Sonar images are noisy, low-contrast, and show heavy class imbalance (up to **6.9x** between classes). No colour information, no lighting, and highly cluttered scenes make standard object detectors unreliable without targeted preprocessing and loss design.

3. Dataset â■■ FLS Marine Debris

ARIS Explorer 3000 sonar frames from three environments. All images: grayscale PNG, 320x480 px.

Sub-dataset	Images	Classes	Annotation	Task
Watertank-Cropped	1,955	10	Folder name	Classification
Turntable-Cropped	4,892	18	Folder name	Classification
Watertank-Segmentation	1,868	11+bg	PNG masks + XML boxes	Detection / Segmentation
Quarry-Fullsize	7,209	—	None	Domain pretraining (opt.)

Watertank-Cropped â■■ 10 Classes

Class	Count	Class	Count	Class	Count
Bottle	449	Hook	133	Shampoo-bottle	99
Can	367	Propeller	137	Standing-bottle	65
Chain	226	Tire	331	Valve	208
Drink-carton	349				

Imbalance: 449 / 65 = **6.9x** | Turntable imbalance: 480 / 176 = 2.7x

Watertank-Segmentation â■■ Pixel Labels

Val	Class	Val	Class	Val	Class
0	Background	4	Drink-carton	8	Standing-bottle
1	Bottle	5	Hook	9	Tire
2	Can	6	Propeller	10	Valve
3	Chain	7	Shampoo-bottle	11	Wall (non-debris)

4. Methodology â■■ Phase by Phase

■ Phase 0 â■■ Environment Setup

• conda env marine-debris: Python 3.11, PyTorch 2.12, MPS backend verified
• ultralytics, transformers, albumentations, pycocotools, OpenCV installed
• Focal loss sanity-checked; MPS device detection confirmed working

■ Phase 1 â■■ Data Inspection & Dataset Classes

• Discovered actual format: PNG + XML bbox + PNG pixel masks (not HDF5)
• FLSClassificationDataset: class-named subfolders, 70/15/15 stratified split
• FLSDetectionDataset: Images/ + XML (x,y,w,h) → normalized bbox tensors
• FLSSegmentationDataset: Images/ + Masks/ → integer label tensors
• Smoke test passed: 1,654 train / 300 val / 301 test samples loaded correctly

■ Phase 2 â■■ Exploratory Data Analysis

• Class distribution histograms for all sub-datasets
• CLAHE effect visualised side-by-side with raw sonar images
• Sample image grid per class with pixel mask overlay
• Imbalance ratios confirm need for weighted sampling and focal loss

■ Phase 3A â■■ Sonar Classification

• ResNet-50 (ImageNet) fine-tuned on Watertank-Cropped (10-class)
• ResNet-50 on Turntable-Cropped (18-class) separately
• Focal loss + WeightedRandomSampler + cosine LR decay
• Target: top-1 accuracy > 75% (Watertank), > 80% (Turntable)

■ Phase 3B â■■ Sonar Detection

• YOLOv8m on Watertank-Segmentation bounding box annotations
• CLAHE → 640x640 → 3-channel replicate; mosaic + HSV augmentation
• Focal loss (fl_gamma=2.0); 11 debris classes (background/wall excluded)
• Target: mAP50 > 35%

■ Phase 4 â■■ Segmentation

• SegFormer-B2 (ADE20K pretrained) on pixel masks
• Combined Focal + Dice loss (0.5/0.5); wall class excluded from mIoU
• Target: mean IoU > 30% across 10 debris classes

■ Phase 5 â■■ Fusion Comparison

• Exp A â■■ Transfer: pretrain ResNet on Turntable (18 cls) → finetune Watertank (10 cls)
• Exp B â■■ Ensemble: YOLO confidence + SegFormer mask probability; sweep α
• Compare vs single-domain baseline to quantify cross-domain benefit

■ Phase 6 â■■ Optical Comparison (Pending)

• Requires TrashCan 1.0 download (5,700 optical images, 3 classes)
• YOLOv8m trained on optical data for direct sonar vs optical comparison
• Final 3-way table: optical-only / sonar-only / fused

5. Preprocessing Pipeline

Every sonar image goes through this fixed pipeline before entering any model:

Step	Parameters	Reason
CLAHE	clip=2.0, tile=8x8	Boosts local contrast in low-SNR sonar
Gaussian blur	sigma=1.0, 3x3 kernel	Suppresses speckle noise
Normalize [0,1]	divide by 255	No ImageNet mean subtraction for sonar
3-channel replicate	stack grayscale x3	Pretrained RGB backbones need 3 channels
Resize	224px (cls) / 640px (det)	Model input requirement
Nearest-neighbour resize	masks only	Preserves integer label values in masks

6. Class Imbalance Strategy

Technique	Applied To	Effect
Focal Loss (alpha=0.25, gamma=2)	All tasks	Down-weights easy majority samples
WeightedRandomSampler	Classification + Seg.	Oversamples minority classes each epoch
Mosaic augmentation	Detection (YOLOv8)	Creates synthetic rare-class compositions
Dice Loss (weight=0.5)	Segmentation only	Optimises overlap for small pixel classes

7. Results Table (Targets)

Experiment	Model	Metric	Target	Status
Watertank classification	ResNet-50	Acc@1	> 75%	■ Pending
Turntable classification	ResNet-50	Acc@1	> 80%	■ Pending
Object detection	YOLOv8m	mAP50	> 35%	■ Pending
Pixel segmentation	SegFormer-B2	mIoU	> 30%	■ Pending
Cross-domain transfer	ResNet-50	mAP50	> baseline	■ Pending
Decision ensemble	YOLO+SegFormer	mAP50	> single	■ Pending
Optical (future)	YOLOv8m	mAP50	> 40%	■ Blocked